

# EXPLORATORY DATA ANALYSIS OF YOUTUBE TRENDING VIDEOS DATASET

Content Trends, Viewership  
Dynamics & User Engagement

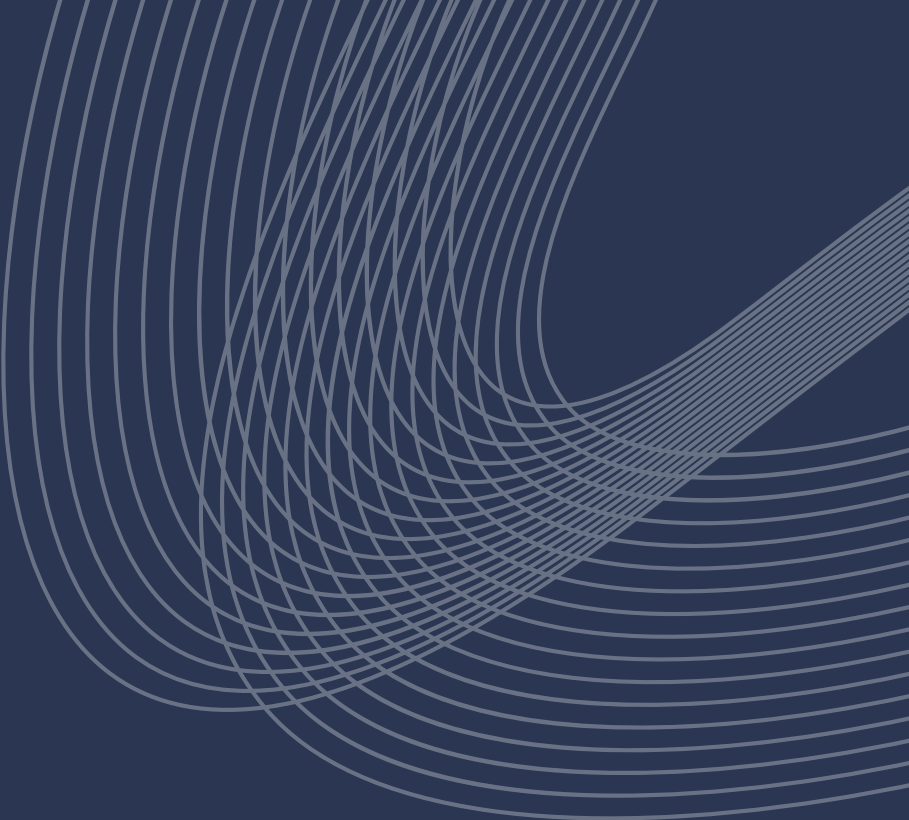


## TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

## PRESENTED BY

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# PROBLEM STATEMENT & OBJECTIVE

## Problem Statement:

The YouTube platform hosts millions of videos across diverse categories. With massive variations in views, likes, and engagement, it becomes challenging to understand content trends, audience preferences, and what drives video success.

## Objective:

To explore the YouTube trending videos dataset and uncover patterns, insights, and trends that help creators and businesses make data driven decisions.



# EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

## Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

## Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

## Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

## Exploratory Analysis

- Univariate Analysis: e.g., Ratings, Installs, Prices
- Bivariate Analysis: Rating vs Reviews
- Multivariate Analysis: Reviews vs Rating vs Installs

05

## Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

# KEY QUESTIONS EXPLORED

- Which categories dominate the Play Store, and where is competition highest?
- What is the distribution of Free vs Paid apps across categories and ratings?
- Who are the target audiences of apps (content rating distribution)?
- How do app sizes vary, and what is the optimal size range?
- How do app ratings behave across categories, review volumes, and time?
- What monetization models work best by category and audience?



# DATA OVERVIEW

The dataset provides key information about YouTube trending videos, including views, likes, comments, category, publish date, and trending date. Additional details such as channel information and country capture content performance, audience engagement and regional trends.

Data Source: Kaggle

## Dataset Size

161470

Records

18

Features

## App Diversity

56905

Videos covered

4

Countries

18

Video Categories



# DATA OVERVIEW

Below is a detailed description of the feature set:

| Dataset Features       | Type                 | Feature Description                                   |
|------------------------|----------------------|-------------------------------------------------------|
| Index                  | Numerical (Discrete) | Dataset row number                                    |
| Video_id               | String               | Unique YouTube video ID                               |
| Trending_date          | Date                 | Date the video trended on YouTube                     |
| Title                  | String               | Video title                                           |
| Channel_title          | String               | Publishing channel name                               |
| Category_id            | Numerical (Discrete) | YouTube category ID                                   |
| Publish_date           | Date                 | Video publication date                                |
| Time_frame             | Categorical          | Price of the app (0 indicates a free app)             |
| Published_day_of_week  | Date                 | Day of the week published                             |
| Publish_country        | Categorical          | Country of publication                                |
| Tags                   | String               | The tags or keywords associated with the video.       |
| Views                  | Categorical          | The number of views received by a particular video    |
| Likes                  | Numerical (Discrete) | The number of likes received by a particular video    |
| Dislikes               | Numerical (Discrete) | The number of dislikes received by a particular video |
| Comment_count          | Numerical (Discrete) | The number of comments received by a particular video |
| Comments_disabled      | Boolean              | Whether comments are disabled                         |
| Ratings_disabled       | Boolean              | Whether ratings are disabled                          |
| Video_error_or_removed | Boolean              | Whether the video is unavailable, removed, or errored |

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# DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

## DATA ANOMALIES

- **Columns requiring cleaning or type conversion:** trending\_date, publish\_date
- **Columns with notable anomalies or invalid entries:**
  - tags (very long strings, |-separated values)
  - publish\_country (may require standardization with short forms)





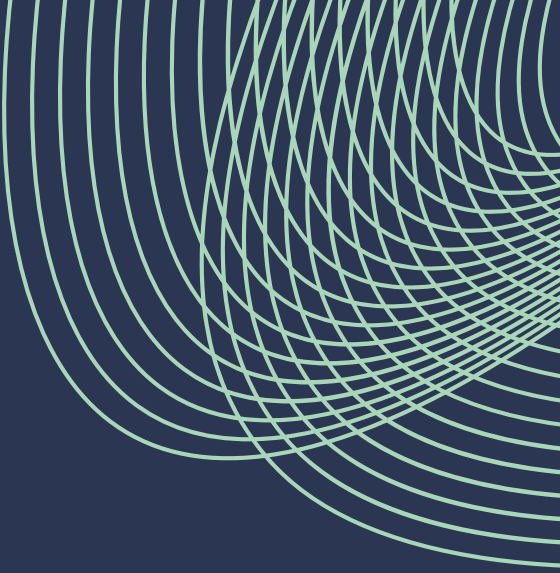
# DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

## DATA CLEANING SUMMARY

- Features such as **trending\_date**, **publish\_date** were standardized with respect to data type conversion.
- Short forms the **publish\_country** feature were standardized.
- The feature **tags** was transformed into a clean, standardized strings by splitting, normalizing, deduplicating and removing irrelevant tags.
- New features **days\_to\_trend** & **category\_name** were created from **trending\_date**, **publish\_date** & **category\_id** respectively.
- **index** and **video\_id** was removed as it is a technical detail with limited business relevance.





# INSIGHTS



# CHALLENGES & OPPORTUNITIES

## Limitations

- The dataset is time bounded and may not reflect current trends or changes.
- Trending videos represent a small subset of all content, creating bias toward highly popular videos.
- Geographic coverage is limited to selected countries and may not capture global viewing behavior.
- Temporal relevance is constrained, as long-term performance and post-trending behavior are not captured.

## Future Work / Opportunities:

- Perform sentiment analysis on video comments to understand audience perception and reactions.
- Conduct time-series analysis on views and likes to study how engagement evolves before and after trending.
- Compare trending videos with non-trending videos to identify factors driving trend eligibility.
- Analyze the impact of video duration, title keywords and thumbnails on engagement and reach.



# CONCLUSION

This analysis of YouTube trending videos provided a comprehensive understanding of content trends, audience behaviour & performance characteristics. Key patterns and opportunities were identified, demonstrating how data driven insights can inform strategic decisions for content creators, marketers, and platforms. Future analysis could include sentiment analysis of viewer comments, time series tracking of engagement patterns, and comparisons with non trending or cross platform video data to uncover deeper insights.

*This analysis was conducted using Python, Pandas, Matplotlib, Seaborn and Plotly.*

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